SOLUTIONS: CONTROL SYSTEMS – 2024/2025

1. Solution

a) Consider the two blocks shown in the block scheme in Figure 1.1 in the text of the exam paper with outputs $r(t)$ and $v(t)$, and transfer functions $\dfrac{6}{s-3}$ and $\dfrac{7}{s-3}$, respectively. Clearly only these blocks represent dynamic systems. Therefore, a *possible choice* for the state variables is:

$$x_1(t) := r(t); \quad x_2(t) := v(t)$$

Hence, we define the state vector:

$$x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Thus, by inspecting the block scheme in Figure 1.1 in the text of the exam paper, we obtain:

$$\begin{cases} \dot{x}_1(t) = 3x_1(t) + 6e(t) \\ \dot{x}_2(t) = 3x_2(t) + 7f(t) \\ e(t) = u(t) + 2y(t) \\ f(t) = e(t) - (2K+1)x_2(t) \\ y(t) = -x_1(t) - Kx_2(t) - u(t) \end{cases}$$

After some trivial algebra, we get:

$$\begin{cases} \dot{x}_1(t) = -9x_1(t) - 12Kx_2(t) - 6u(t) \\ \dot{x}_2(t) = -14x_1(t) - 4(1+7K)x_2(t) - 7u(t) \\ y(t) = -x_1(t) - Kx_2(t) - u(t) \end{cases}$$

Accordingly, the matrices A, B, C, D of the state space description are given by:

$$A = \begin{bmatrix} -9 & -12K \\ -14 & -4(1+7K) \end{bmatrix}, \; B = \begin{bmatrix} -6 \\ -7 \end{bmatrix}, \; C = \begin{bmatrix} -1 & -K \end{bmatrix}, \; D = -1$$

[7 Marks]

b) The stability properties of the dynamic system shown in Figure 1.1 of the exam paper as a function of the scalar $K \in \mathbb{R}$ only depend on the state matrix A of any state-space description of the system. From the answer to Question 1-a), we have:

$$A = \begin{bmatrix} -9 & -12K \\ -14 & -4(1+7K) \end{bmatrix}$$

which **does depend on** K. Therefore, the stability properties also do depend on K. The characteristic polynomial is:

$$\phi_A(s) = \det(sI - A) = (s+9)[s+4(1+7K)] - 12 \cdot 14 \cdot K = s^2 + (13+28K)s + (36+84K)$$

Thus, it is immediate to conclude that the system is asymptotically stable if $K > -36/84$, it is stable (marginally) if $K = -36/84$, and it is unstable if $K < -36/84$.

[3 Marks]

Answers

c) If we set $K = -1/2$, the block scheme in Figure 1.1 in the text of the exam paper reduces to the following block scheme:

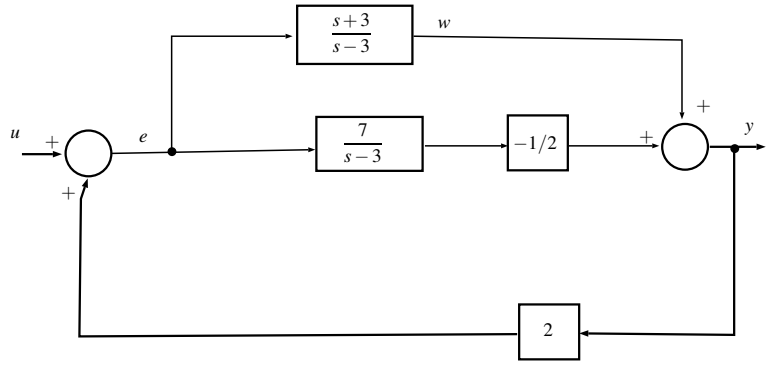


Figure 1.1 Reduced block scheme for $K = -1/2$.

By inspecting the above reduced block scheme, it turns out immediately that there are two blocks in parallel in the direct path and hence we get the further reduced block scheme in figure below:

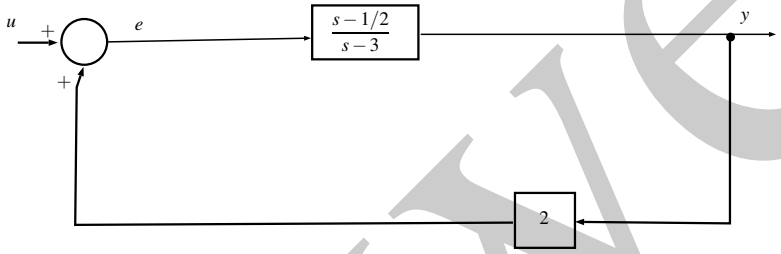


Figure 1.2 Further reduced block scheme for $K = -1/2$.

By inspecting the block scheme in Fig. 1.2, it follows that one pole in $s = -3$ has "disappeared" which implies that there is a "unstable hidden dynamics" and hence the overall system for $K = -1/2$ is unstable as well.

Since $K = -1/2 < -36/84$, the above conclusion is consistent with the analysis reported in the answer to Question 1-b).

[10 Marks]

d) Setting $K = 0$, the block scheme in Figure 1.1 in the text of the exam paper reduces to:

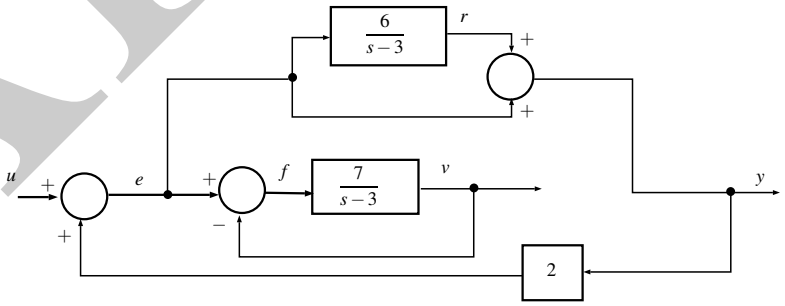


Figure 1.3 Reduced block scheme for $K = 0$.

By simple calculations owing to the standard rules of block algebra, one finally

Answers

obtains:

$$G_{u,y}(s) = \frac{\dfrac{s+3}{s-3}}{1-2\dfrac{s+3}{s-3}} = -\frac{s+3}{s+9}$$

[10 Marks]

e) Setting $K = 1$, the block scheme in Figure 1.1 in the text of the exam paper reduces to:

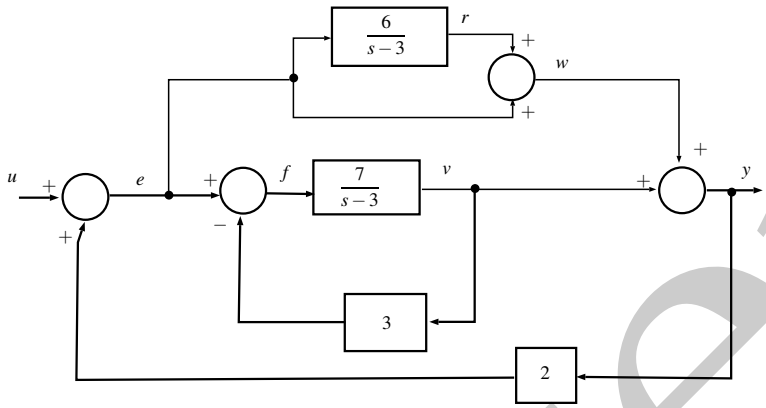


Figure 1.4 Reduced block scheme for $K = 1$.

Hence, owing to the standard rules of block algebra, we obtain:

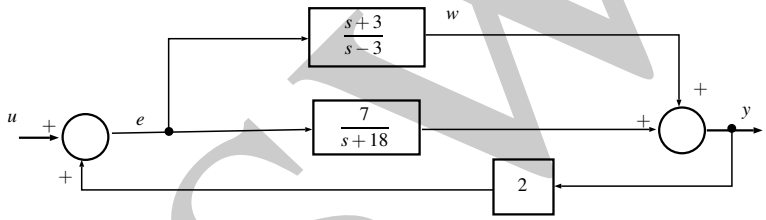


Figure 1.5 Further reduced block scheme for $K = 1$.

Finally, again owing to the standard rules of block algebra, after some algebraic calculations,

$$G_{u,y}(s) = \frac{\dfrac{s+3}{s-3}+\dfrac{7}{s+18}}{1-2\left(\dfrac{s+3}{s-3}+\dfrac{7}{s+18}\right)} = \cdots = -\frac{s^2+28s+33}{s^2+41s+120}$$

[10 Marks]

Answers

2. Solution

a) i) To determine all possible equilibrium states $\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$ corresponding to the constant input $u(t) = \bar{u} = 3, \forall t$, we impose the following conditions:

$$\begin{cases} 0 &= 4[\bar{x}_1]^4 + 2[\bar{x}_1]^2 + 2\bar{x}_2 - 3 \\ 0 &= -2[\bar{x}_1]^4 - \bar{x}_2 - 7.5 \end{cases}$$

By simple algebra, we obtain:

$$\begin{cases} \bar{x}_2 &= -\left(2\bar{x}_1^4 + 7.5\right) \\ \bar{x}_1^2 &= 9 \end{cases}$$

Therefore, for the constant input $u(t) = \bar{u} = 3, \forall t$, we obtain two equilibrium states:

$$\bar{x} = \begin{bmatrix} 3 \\ -169.5 \end{bmatrix}; \qquad \widetilde{x} = \begin{bmatrix} -3 \\ -169.5 \end{bmatrix}$$

[3 Marks]

ii) Now, we consider the equilibrium state $\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ corresponding to the constant input $u(t) = 0, \forall t$. From the nonlinear system

$$\begin{cases} \dot{x}_1(t) &= 4[x_1(t)]^4 + 2[x_1(t)]^2 + 2x_2(t) - u(t) \\ \dot{x}_2(t) &= -2[x_1(t)]^4 - x_2(t) - 2.5u(t) \end{cases}$$

given in the text of Question 2-a), the state and input matrices A and B of the linearised system on the equilibrium state $\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ are given by, respectively:

$$A = \left.\frac{\partial f}{\partial x}\right|_{\bar{x},\bar{u}} = \left[\begin{array}{c|c} 16x_1^3 + 4x_1 & 2 \\ \hline -8x_1^3 & -1 \end{array}\right]_{\bar{x},\bar{u}} = \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\bar{x},\bar{u}} = \begin{bmatrix} -1 \\ -2.5 \end{bmatrix}_{\bar{x},\bar{u}} = \begin{bmatrix} -1 \\ -2.5 \end{bmatrix}$$

[3 Marks]

iii) Since $A = \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix} \Longrightarrow$

$$\begin{aligned} \det(sI - A) &= \det \begin{bmatrix} s & -2 \\ 0 & s+1 \end{bmatrix} \\ &= s(s+1) \end{aligned}$$

Thus, the eigenvalues are the roots of $\det(sI - A)$: $\lambda_1 = -1 < 0$, $\lambda_2 = 0 > 0$. Since $\lambda_2 = 0$, no conclusions can be drawm on the stability of the equilibrium state $\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

[3 Marks]

Answers

b) Consider the state equations given in the text of Question 2-b):

$$\begin{cases} \dot{x}_1(t) &= -x_1(t) + \dfrac{1}{2}\, x_2(t) + u(t) \\ \dot{x}_2(t) &= -2x_1(t) + x_2(t) \\ y(t) &= -x_2(t) \end{cases}$$

with the initial state set to $x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and the input set as $u(t) = \mathbf{1}(t)$, where $\mathbf{1}(t)$ denotes the unit step function.

From the table of the Laplace transforms, we get:

$$U(s) = \frac{1}{s}$$

The matrices A, B, C, D of the state space description of the above linear system are given by:

$$A = \begin{bmatrix} -1 & 1/2 \\ -2 & 1 \end{bmatrix},\ B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},\ C = \begin{bmatrix} 0 & -1 \end{bmatrix},\ D = 0$$

The Laplace transform of the output function $y(t)$ is given by the general formula:

$$Y(s) = C(sI - A)^{-1}x(0) + C(sI - A)^{-1}BU(s)$$

We get:

$$sI - A = \begin{bmatrix} s+1 & -1/2 \\ 2 & s-1 \end{bmatrix} \quad \Longrightarrow \quad \det(sI - A) = s^2 - 1 + 1 = s^2$$

Hence:

$$(sI - A)^{-1} = \frac{1}{s^2} \begin{bmatrix} s-1 & 1/2 \\ -2 & s+1 \end{bmatrix}$$

Therefore:

$$C(sI - A)^{-1}x(0) = \frac{1}{s^2} \begin{bmatrix} 0 & -1 \end{bmatrix} \begin{bmatrix} s-1 & 1/2 \\ -2 & s+1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \cdots = -\frac{s+1}{s^2} = -\frac{1}{s} - \frac{1}{s^2}$$

and

$$C(sI - A)^{-1}BU(s) = \frac{1}{s^2} \begin{bmatrix} 0 & -1 \end{bmatrix} \begin{bmatrix} s-1 & 1/2 \\ -2 & s+1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{1}{s} = \cdots = \frac{2}{s^3}$$

Using the above results and inspecting the table of the Laplace transforms, it follows that:

$$y(t) = \mathscr{L}^{-1}[Y(s)] = \mathscr{L}^{-1}\left[-\frac{1}{s} - \frac{1}{s^2} + \frac{2}{s^3} \right] = [t^2 - t - 1] \cdot \mathbf{1}(t)$$

[11 Marks]

Answers

3. Solution

a) First, it is convenient to separate the effect of the delay T by writing:

$$G(s) = \widetilde{G}(s) \cdot e^{-Ts} \quad \text{with} \quad \widetilde{G}(s) = 4.8 \cdot \frac{1+0.15s}{1-0.15s}$$

Since the system represented by $\widetilde{G}(s)$ is open-loop unstable, the Bode stability criterion cannot be used in the way that is usual for the loop-shaping approach to the design of the controller for minimum-phase systems. Hence, as suggested in Example 3 in slides 34-37 of Part 10 of the lecture handouts, we use a dual-loop control scheme shown in Fig. 3.1 below that has a different structure compared with the standard feedback scheme shown in Fig. 3.1 in the Questions part and where $C_1(s)$ is designed so as the inner-loop transfer function $F_1(s)$ is asymptotically stable. Subsequently, $C_2(s)$ will be designed to meet the other requirements.

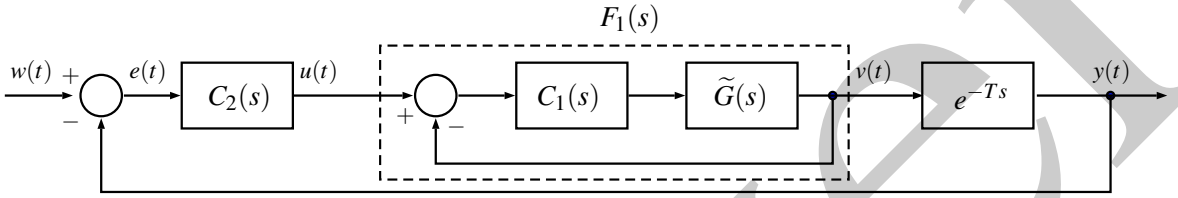


Figure 3.1 The dual-loop control structure for Question 3.

According to the specifications in the text of the exam paper, we set $T = 0$ seconds. As a first attempt to obtain an asymptotically stable inner control loop (that is, the poles of the transfer function $F_1(s)$ from the input $u(t)$ to the output $v(t)$ are all located in the open left half-plane $\mathbb{C}^-$ of the complex plane $\mathbb{C}$), we set $C_1(s) = \mu_1 \in \mathbb{R}$.

With this choice of $C_1(s)$, let us calculate the transfer function $F_1(s)$:

$$F_1(s) = \frac{\mu_1 \cdot 4.8 \cdot \dfrac{1+0.15s}{1-0.15s}}{1+\mu_1 \cdot 4.8 \cdot \dfrac{1+0.15s}{1-0.15s}} = \cdots = \frac{4.8 \cdot \mu_1 \cdot (1+0.15s)}{\left(\mu_1 \cdot \dfrac{18}{25} - \dfrac{15}{100}\right) s + 1 + \dfrac{48}{10} \cdot \mu_1}$$

The closed-loop pole of $F_1(s)$ is given by

$$p = -\frac{1+\dfrac{48}{10} \cdot \mu_1}{\mu_1 \cdot \dfrac{18}{25} - \dfrac{15}{100}}.$$

Therefore, setting $\mu_1 \in \mathbb{R}$ such that $|\mu_1| > 5/24$ ensures that $p < 0$.

We choose (for example) $\mu_1 = -115/408 \simeq -0.28$, which gives $p = -1$. Thus, simple algebra gives:

$$F_1(s) = \frac{23}{6} \cdot \frac{1+\dfrac{3}{20}s}{1+s}$$

Since $F_1(s)$ is minimum phase, in the case $T = 0$ set in Question 3-a), the remaining design of the controller $C_2(s)$ can be carried out via the usual loop-shaping approach.

Answers

To get insight into the constraints that Requirements i) and ii) impose in the frequency domain, let us sketch the straight-line approximation of the Bode diagram and a few points of the true Bode diagram of open-loop transfer function $L(s)$ setting, for the time being, $C_2(s) = \mu_2 = 1$. Hence:

$$L(s) = \mu_2 F_1(s) = \frac{23}{6} \cdot \frac{1 + \frac{3}{20}s}{1+s}$$

The parameters of $L(s)$ for $\mu_2 = 1$ are:

$$\mu = 23/6, \quad g = 0 \text{ (no poles in } s = 0),$$

and the relevant frequencies are:

$$\omega_{z_1} = 20/3 \simeq 6.67, \quad \omega_{p_1} = 1\,.$$

- When $0 < \omega < 1$, the slope is 0 dB$/$dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$|L|_{\text{dB}} = 20\log(23/6) \simeq 11.7\text{dB}.$$

- When $1 \leq \omega \leq 20/3$, the slope is -20 dB$/$dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$\begin{aligned} |L|_{\text{dB}} &= -20\log\omega + \mu|_{\text{dB}} \\ &= -20\log\omega + 20\log(23/6) \simeq -20\log\omega + 11.7\text{dB}\,. \end{aligned}$$

  For $\omega = 20/3$, the magnitude is

$$|L|_{\text{dB}} \simeq -20\log(20/3) + 20\log(23/6) \simeq -4.8\text{dB}.$$

- When $\omega > 20/3$, the slope is 0dB$/$dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$|L|_{\text{dB}} \simeq -4.8\text{dB}.$$

Concerning the phase, we write:

$$\arg L(j\omega) = \arg(1 + \frac{3}{20}j\omega) - \arg(1 + j\omega)$$

We calculate $\arg L(j\omega)$ for $\omega = 1, 20/3$ thus getting:

$$\arg L(j) \simeq -36.5^\circ; \arg L(j20/3) \simeq -36.5^\circ$$

The resulting true Bode diagrams with their straight-line approximation are plotted in Figure 3.2 below.

Owing to Requirement iii), the controller has to be strictly proper and the current choice for $C_2(s) = \mu_2$ does not meet this requirement.

Since $F_1(s)$ is asymptotically stable and minimum phase, proceeding by redesigning the controller to meet Requirements i)–iii) via the "cancellation" of the open-loop poles and zeros of $F_1(s)$ is feasible.

Hence, we set:

$$\widetilde{C}_2(s) = \widetilde{\mu}_2 \frac{23}{6} \cdot \frac{1+s}{1 + \frac{3}{20}s} \cdot \frac{1}{1+s/20}$$

Answers

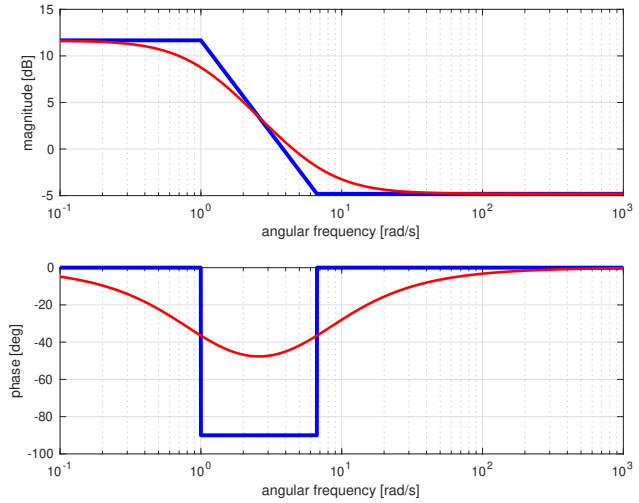


Figure 3.2 Bode diagrams with $\mu_2 = 1$.

where the additional term $\dfrac{1}{1+s/20}$ is added in order to meet Requirement iii) and to introduce a high-frequency pole in order to possibly meet Requirements i)–ii) as well by properly designing the gain $\widetilde{\mu}_2$.

For the time-being, let's set $\widetilde{\mu}_2 = 1$ Hence, the new open-loop transfer function reads:

$$\widetilde{L}(s) = \widetilde{\mu}_2 F_1(s)\cdot \widetilde{C}_2(s) = \frac{23}{6}\cdot\frac{1+\dfrac{3}{20}s}{1+s}\cdot\frac{1+s}{1+\dfrac{3}{20}s}\cdot\frac{1}{1+s/20} = \frac{23}{6}\cdot\frac{1}{1+s/20}$$

The parameters of $\widetilde{L}(s)$ for $\widetilde{\mu}_2 = 1$ are:

$$\mu = 23/6, \quad g = 0 \text{ (no poles in } s = 0),$$

and the relevant frequency is:

$$\omega_{p_1} = 20\,.$$

Let us now sketch the straight-line approximation of the Bode diagram and a few points of the true Bode diagram of open-loop transfer function $\widetilde{L}(s)$:

- When $0 < \omega \leq 20$, the slope is 0 dB/dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$|\widetilde{L}|_{\text{dB}} = 20\log(23/6) \simeq 11.7\text{dB}.$$

- When $\omega > 20$, the slope is -20 dB/dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$\begin{aligned}|\widetilde{L}|_{\text{dB}} &= -20(\log\omega - \log 20) + \mu|_{\text{dB}} \\ &= -20\log\omega + 20\log(20) + 20\log(23/6) \simeq -20\log\omega + 37.7\text{dB}\,.\end{aligned}$$

For $\omega = 50$, the magnitude is

$$|\widetilde{L}|_{\text{dB}} \simeq -20\log(50) + 20\log(20) + 20\log(23/6) \simeq 3.7\text{dB}.$$

Answers

Concerning the phase, we write:

$$\arg\widetilde{L}(j\omega) = -\arg(1 + j\omega/20)$$

We calculate $\arg L(j\omega)$ for $\omega = 20$ thus getting:

$$\arg\widetilde{L}(j20) = 45^\circ$$

The resulting true Bode diagrams with their straight-line approximation are plotted in Figure 3.3 below.

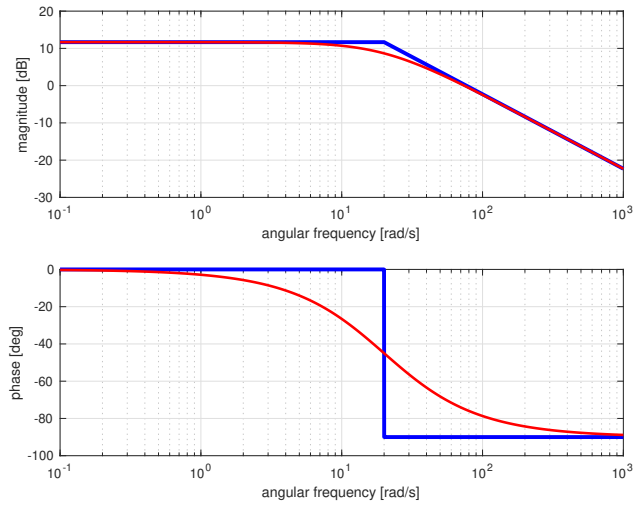


Figure 3.3 Bode diagrams with $\widetilde{\mu}_2 = 1$.

In order to meet Requirement i), we set the crossover frequency as $\omega_c = 15\,\text{rad/seconds}$ and we calculate the corresponding phase margin to check whether Requirement ii) would be satisfied (as it clearly seems from inspection of Figure 3.3) and we calculate the corresponding phase margin:

$$\arg\widetilde{L}(j\omega_c) = \arg\widetilde{L}(15j) = -\arg(1 + \frac{15}{20}j) \simeq -36.9^\circ$$

The phase margin is $\varphi_m \simeq 180^\circ - 36.9^\circ \simeq 143^\circ$ which abundantly satisfies Requirement ii) on the phase margin.

At last, we calculate the value of the gain $\widetilde{\mu}_2$ that sets the crossover frequency to the desired value $\omega_c = 15\,\text{rad/seconds}$.

Specifically, we need to compute the value of the gain $\widetilde{\mu}_2$ such that

$$|\widetilde{L}(j\omega_c)|_{\text{dB}} = |\widetilde{L}(15j)|_{\text{dB}} = 0\text{dB}$$

Hence:

$$20\log\widetilde{(\mu_2)} = 20\log|1 + j15/20| - \log(23/6)$$

which gives

$$20\log(\widetilde{\mu}_2) \simeq -9.73\text{dB} \Longrightarrow \widetilde{\mu}_2 = 10^{-9.73/20} \simeq 0.326$$

Therefore, the controller $\widetilde{C}_2(s)$ is

$$\widetilde{C}_2(s) = 0.326 \cdot \frac{1+s}{1+\frac{3}{20}s} \cdot \frac{1}{1+s/20}$$

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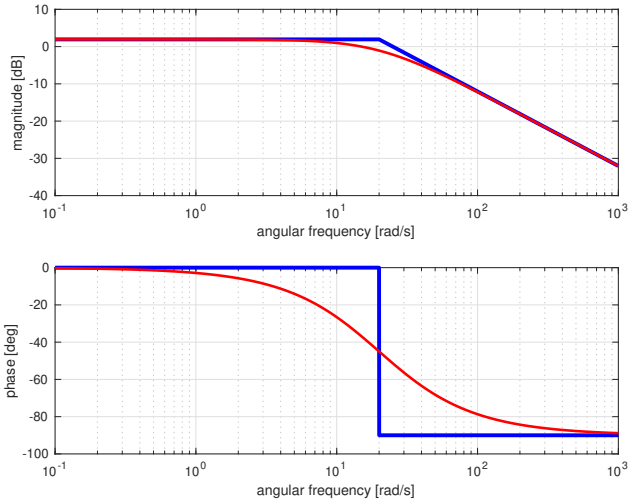


Figure 3.4 Bode diagrams of the frequency response of the open-loop transfer function $\widetilde{L}(s)$ under the action of the final designed controller.

The resulting true Bode diagrams with their straight-line approximation are plotted in Figure 3.4 above.

[20 Marks]

b) i) Using the dual-loop controller obtained in answering Question 3-a), the open-loop transfer function reads:

$$\widetilde{L}(s) = \frac{5}{4} \cdot \frac{1}{1+s/20}$$

The complementary sensitivity function $F(s)$, that is, the transfer function from the input $w(t)$ to the output $y(t)$ is given by:

$$F(s) = \frac{\widetilde{L}(s)}{1+\widetilde{L}(s)} = \frac{\frac{5}{4} \cdot \frac{1}{1+s/20}}{1+\frac{5}{4} \cdot \frac{1}{1+s/20}} = \cdots = \frac{25}{s+45}$$

Hence:

$$Y(s) = \mathscr{L}[y(t)] = F(s) \cdot \mathscr{L}[\left(1-e^{-t}\right) \cdot \mathbf{1}(t)] = \frac{25}{s+45} \cdot \frac{1}{s(s+1)}$$

Proceeding by partial fractions expansion we get:

$$Y(s) = \frac{\alpha}{s} + \frac{\beta}{s+1} + \frac{\gamma}{s+45}$$

where

$$\alpha = 25 \frac{1}{(s+1)(s+45)}\bigg|_{s=0} = \frac{25}{45} = \frac{5}{9}$$

$$\beta = 25 \frac{1}{s(s+45)}\bigg|_{s=-1} = -\frac{25}{44}$$

$$\gamma = 25 \frac{1}{s(s+1)}\bigg|_{s=-45} = \frac{25}{(-45)(-44)} = \frac{5}{396}$$

Answers

Finally:

$$y(t) = \mathscr{L}^{-1}[Y(s)] = \left(\frac{5}{9} - \frac{25}{44}e^{-t} + \frac{5}{396}e^{-45t}\right) \cdot \mathbf{1}(t)$$

[6 Marks]

ii) Consider the reference input function $w(t), t \in \mathbb{R}$ and of the output function $y(t), t \in \mathbb{R}$ determined in the above answer to Question 3-b)-i), that is:

$$w(t) = \left(1 - e^{-t}\right) \cdot \mathbf{1}(t); \quad y(t) = \left(\frac{5}{9} - \frac{25}{44}e^{-t} + \frac{5}{396}e^{-45t}\right) \cdot \mathbf{1}(t).$$

Figure 3.5 below shows the time behaviour of the reference input function $w(t)$ and the corresponding output function $y(t)$.

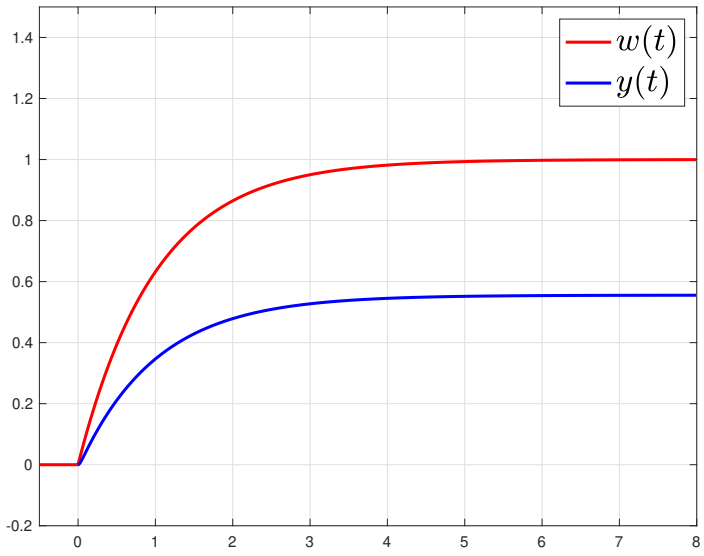


Figure 3.5 Time behaviour of the input $w(t)$ and the output $y(t)$.

[3 Marks]

iii) By comparing the time-behaviours of the reference input function $w(t)$ and the output function $y(t)$ drawn in Figure 3.5, we notice the presence of a significant steady-state error. This is not surprising as no requirements on the static error have been given for the design of the controller.

[3 Marks]

c) According to Question 3-c), let us consider the presence of the delay $T \neq 0$. In particular, suppose that $0 < T \leq \overline{T}$, where $\overline{T} > 0$ is an upper bound on the delay T and let us consider the dual-loop control system with the structure shown in Figure 3.1 of the present Solutions document and with the controllers $C_1(s)$ and $\widetilde{C}_2(s)$ designed in the answer to Question 3-a) for $T = 0$ seconds.

Ans

The presence of a delay $T \neq 0$ causes a decrease $\Delta\varphi_m$ of the phase margin φ_m given by:

$$\Delta\varphi_m = \arg e^{-j\omega_c T}$$

Therefore, a upper bound $\overline{T} > 0$ such that the decrease $\Delta\varphi_m$ of the phase margin φ_m caused by any time-delay T with $0 < T \leq \overline{T}$ satisfies $|\Delta\varphi_m| \leq 20^\circ$ can be computed solving:

$$-\omega_c \overline{T} \frac{180^\circ}{\pi} = -15\overline{T} \frac{180^\circ}{\pi} = -20^\circ \Longrightarrow \overline{T} = \frac{\pi}{180^\circ} \cdot \frac{20}{15} \simeq 0.023 \,\text{seconds}$$

It is worth noting that $\overline{T}$ is rather small which is due to the required rather high value of the crossover frequency.

[8 Marks]

Answers